

Solo E-Commerce Platform — High Level Design (HLD)

Document Version: 1.0

Date: 17 March 2026

Author: Solo Engineering Team

Status: Final

1. Executive Summary

Solo E-Commerce is a premium kitchenware, tableware, and home goods online retail platform built for the UAE market. The platform enables customers to browse, search, and purchase curated products from brands such as Eva Solo, Eva Trio, and PWtbS, with full support for AED currency, VAT compliance (5%), and local delivery logistics.

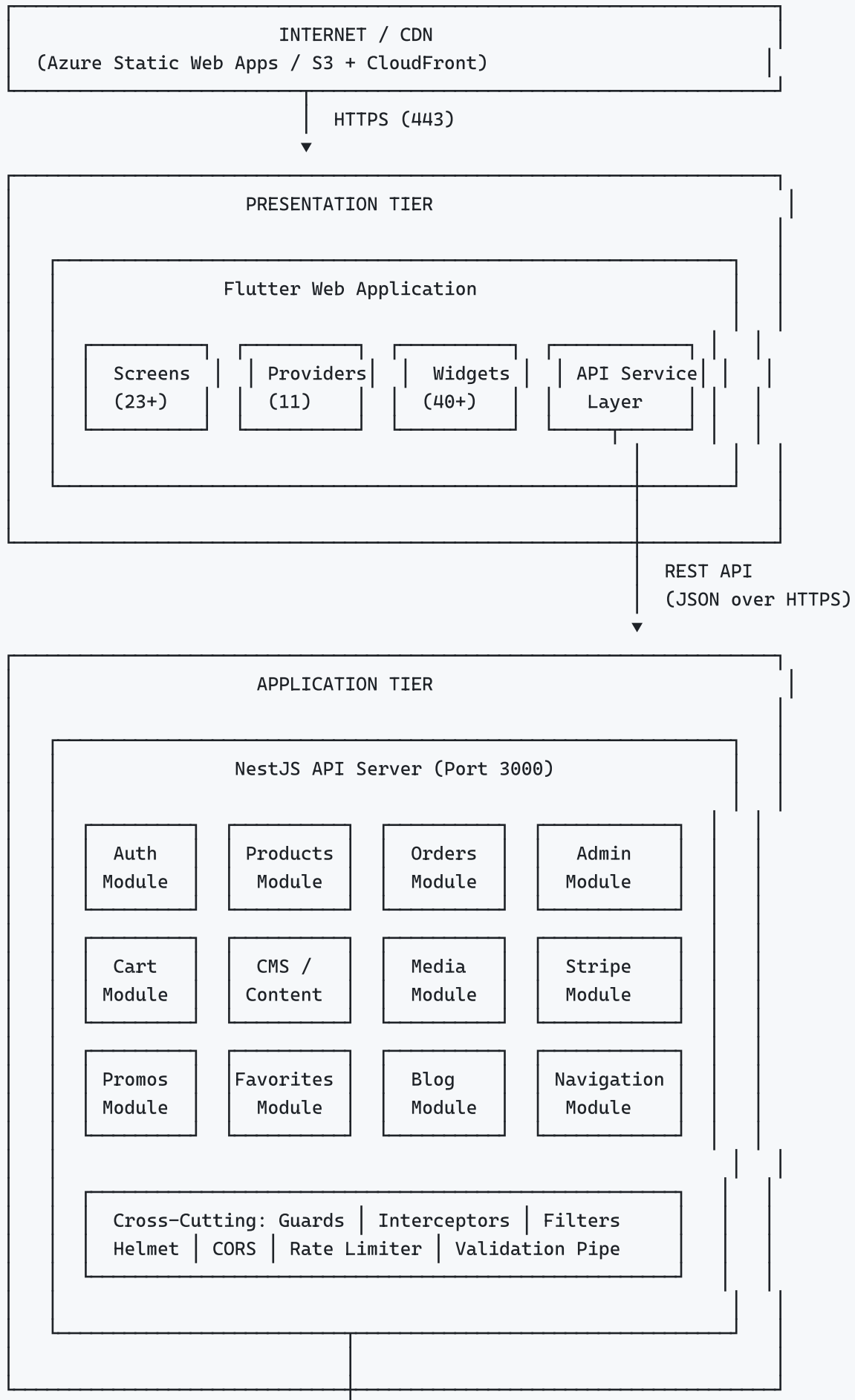
The system is built as a **three-tier web application**: a Flutter Web frontend, a NestJS REST API backend, and a PostgreSQL relational database layer. It supports two distinct user personas — **Customers** (shopping, account management, loyalty) and **Administrators** (inventory, orders, CMS, analytics).

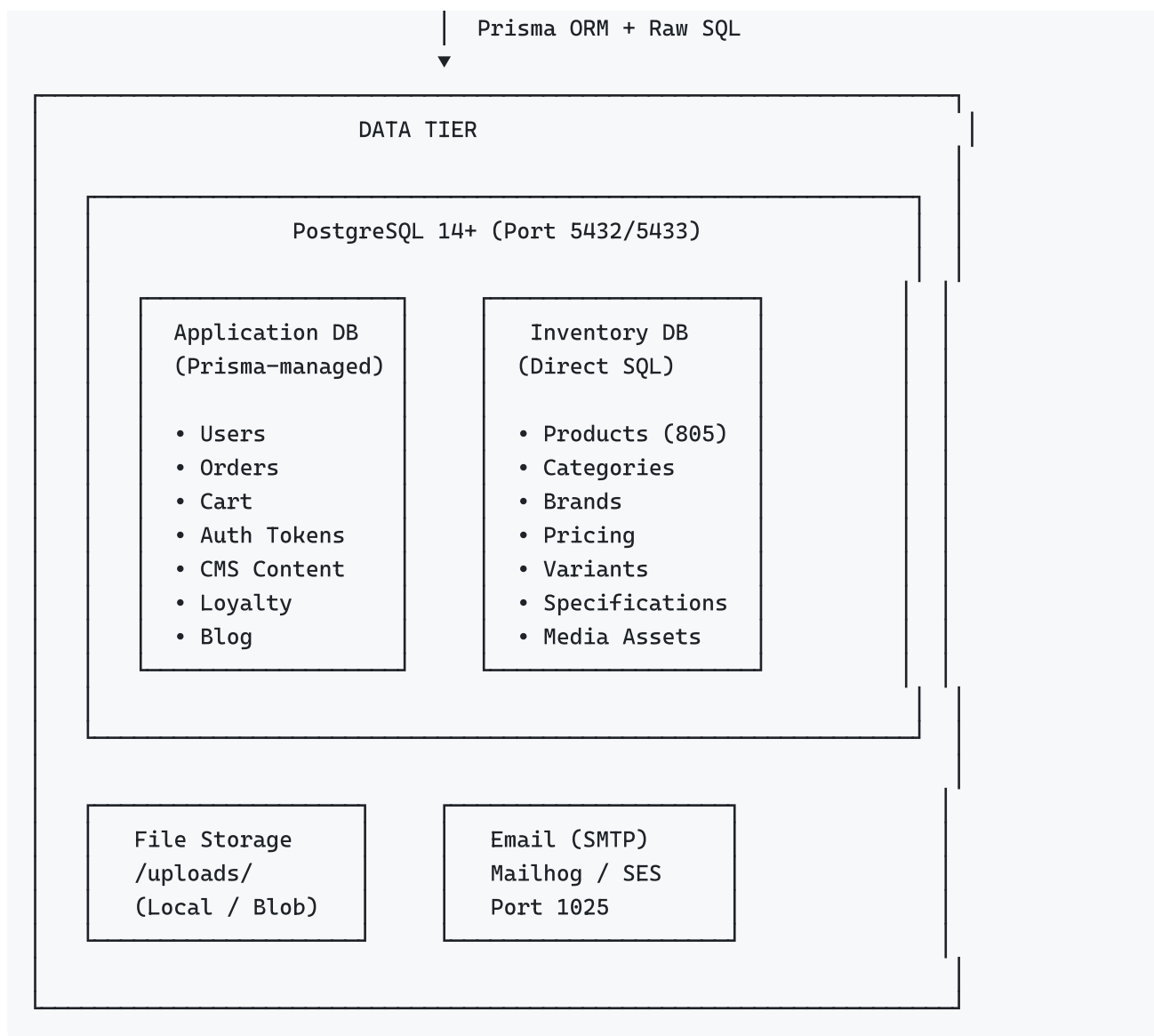
2. System Goals & Objectives

Goal	Description
G1 — Online Retail	Provide a full e-commerce experience: browse → search → cart → checkout → payment → order tracking
G2 — Admin Operations	Enable business staff to manage products, orders, customers, promotions, and content without developer involvement
G3 — Content Management	Allow dynamic homepage configuration, banners, landing pages, and blog content via a built-in CMS
G4 — Payment Processing	Integrate Stripe for secure credit/debit card payments with PCI-DSS compliance
G5 — Loyalty & Retention	Implement a points-based loyalty program to encourage repeat purchases
G6 — Scalability	Design for horizontal scaling — stateless backend, managed database, CDN-served frontend

Goal	Description
G7 — Security	Follow OWASP Top 10 guidelines for authentication, authorization, input validation, and data protection

3. High-Level System Architecture





4. Key Components

4.1 Presentation Tier — Flutter Web Application

Aspect	Detail
Framework	Flutter 3.38+ / Dart 3.10+
Rendering	Material Design 3, responsive breakpoints
State Management	Provider pattern (11 providers)
Routing	Named route navigation with guards
API Communication	Custom ApiClient with JWT interceptor, token refresh
Build Output	Static HTML/JS/CSS bundle served via CDN or static host

4.2 Application Tier — NestJS REST API

Aspect	Detail
Framework	NestJS 10.3 / Node.js 18+ / TypeScript 5.3+
API Style	RESTful JSON, <code>/api</code> prefix, versioned by convention
Modules	21+ domain modules, each self-contained
Security	Helmet, CORS, Throttler (1000 req/min), DTO validation
Auth	JWT (access 15m + refresh 7d), Argon2id password hashing
Payments	Stripe SDK (Payment Intents API)
File Handling	Multer for uploads, local storage with CDN path resolution
Total Endpoints	170+ REST endpoints

4.3 Data Tier — PostgreSQL Database

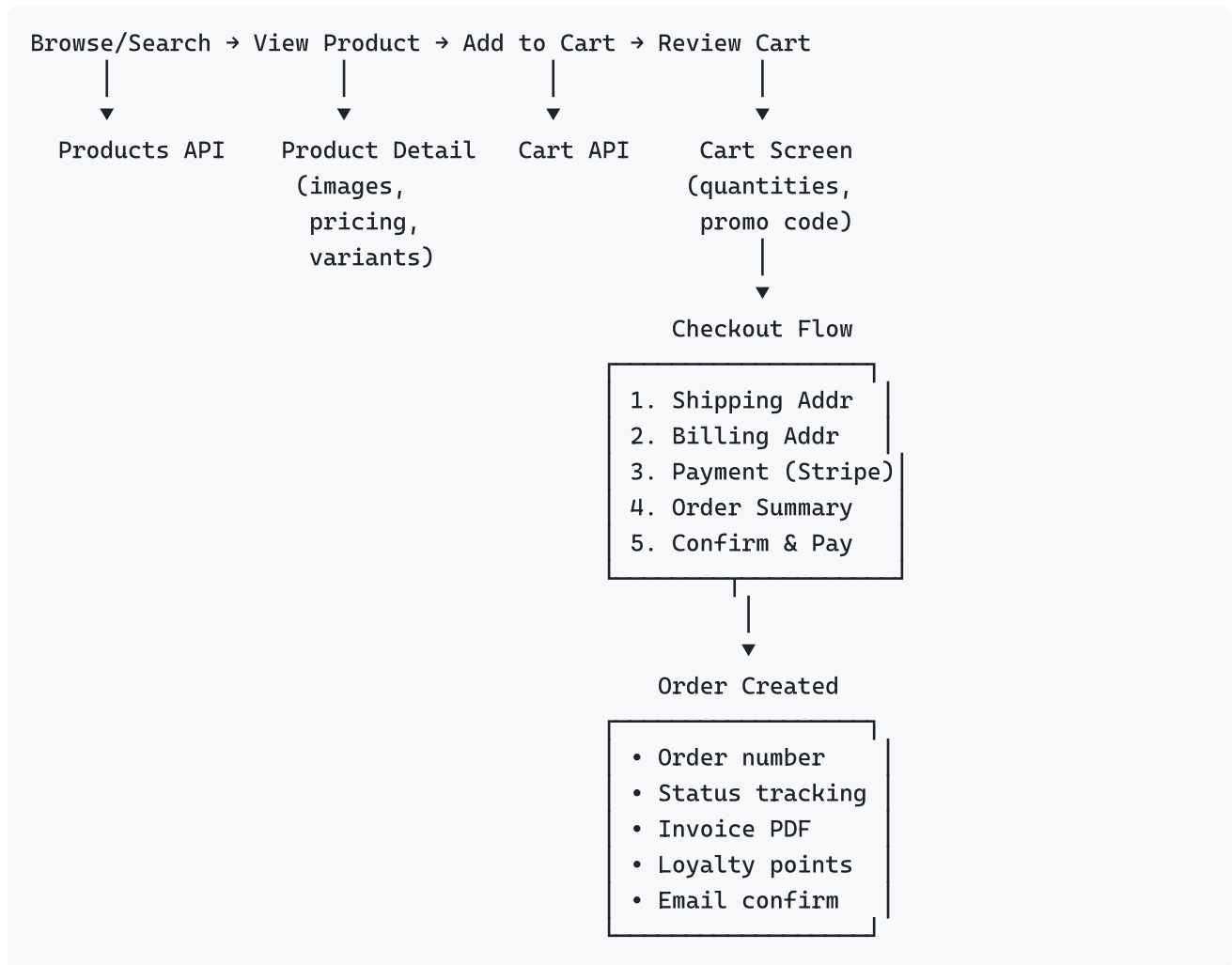
Aspect	Detail
Engine	PostgreSQL 14+
ORM	Prisma 5.22 for application models, raw SQL for inventory
Models	35+ tables across dual-database architecture
Products	805 products, 3 categories, 4 brands
Migrations	Prisma Migrate for schema evolution
Backups	Automated via cloud provider (Azure/AWS)

5. User Personas & Roles

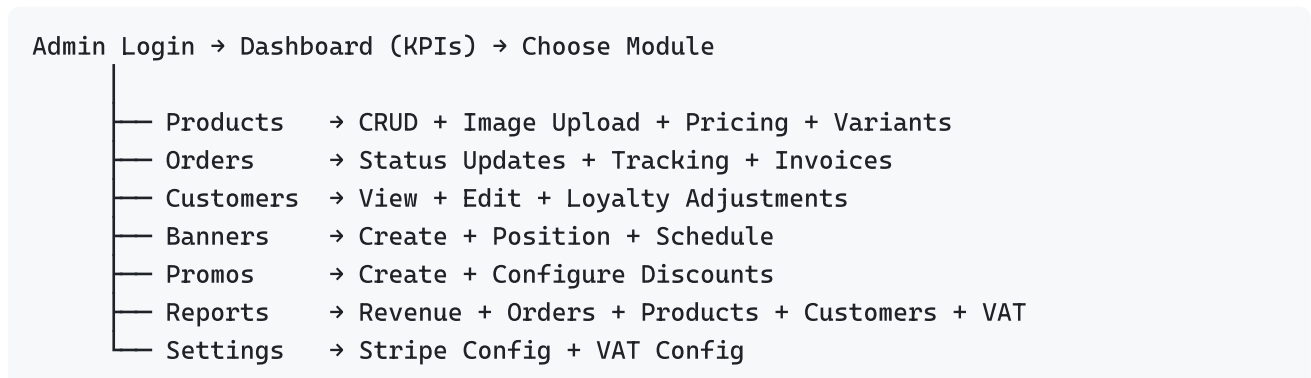
Role	Access Level	Capabilities
Anonymous Visitor	Public endpoints only	Browse products, view categories, read blog, search
Registered Customer	Authenticated (JWT)	Add to cart, checkout, manage account, favorites, loyalty, order history
Admin	ADMIN role required	Manage products, orders, customers, banners, promotions, reports
Super Admin	SUPER_ADMIN role	All Admin + system settings, VAT config, Stripe config

6. Core Business Flows

6.1 Customer Purchase Flow



6.2 Admin Operations Flow



7. Integration Points

Integration	Purpose	Protocol
Stripe	Payment processing (card payments)	HTTPS REST API
SMTP (Mailhog/SES)	Transactional emails (order confirm, password reset)	SMTP (Port 1025 dev / 587 prod)
File Storage	Product images, banners, media assets	Local filesystem / Azure Blob / S3
CDN	Static frontend delivery, image caching	HTTPS (Azure CDN / CloudFront)

8. Non-Functional Requirements

Requirement	Target
Availability	99.9% uptime SLA
Response Time	API p95 < 500ms
Concurrent Users	100+ simultaneous sessions
Data Retention	Orders retained indefinitely; logs retained 90 days
Security	OWASP Top 10 compliance, PCI-DSS via Stripe
Scalability	Horizontal scaling via stateless API + managed DB
Accessibility	WCAG 2.1 AA target
Browser Support	Chrome, Safari, Firefox, Edge (latest 2 versions)

9. Deployment Topology

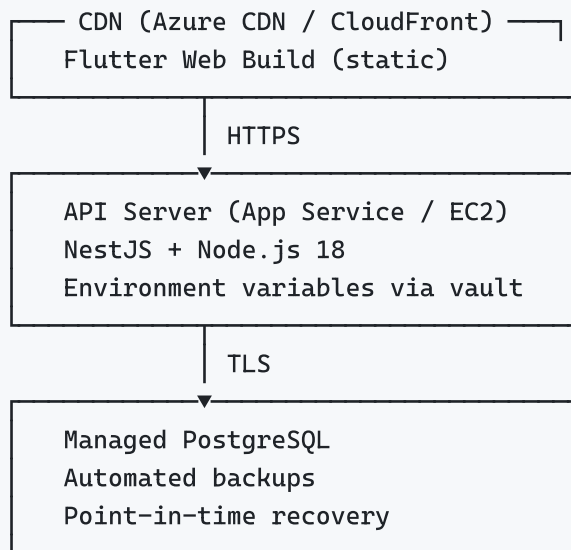
Development Environment

```

Developer Machine
├── Flutter DevTools      → localhost:52391
├── NestJS Dev Server     → localhost:3000
├── PostgreSQL           → localhost:5432/5433
├── Mailhog SMTP          → localhost:1025
└── Mailhog Web UI        → localhost:8025

```

Production Environment (Cloud)



10. Technology Stack Summary

Layer	Technology	Version
Frontend Framework	Flutter Web	3.38+
Frontend Language	Dart	3.10+
State Management	Provider	6.1+
Backend Framework	NestJS	10.3
Backend Language	TypeScript	5.3+
Runtime	Node.js	18+
Database	PostgreSQL	14+
ORM	Prisma	5.22
Authentication	JWT + Argon2id	—
Payments	Stripe SDK	Latest
Email	Nodemailer + SMTP	—
File Upload	Multer	—
API Security	Helmet, Throttler, CORS	—
Containerization	Docker Compose	—
Design System	Material Design 3	—

11. Data Volume Estimates

Entity	Current Count	Growth Projection (1 Year)
Products	805	1,500–2,000
Categories	3	8–12
Brands	4	8–15
Registered Users	~50	500–2,000
Orders/Month	~10	200–500
Media Assets	~2,000	5,000–10,000
Blog Posts	~20	100–200

12. Risk & Mitigation

Risk	Impact	Mitigation
Single server failure	Service downtime	Cloud-managed auto-restart, health checks
Database corruption	Data loss	Automated daily backups, point-in-time recovery
Payment breach	Financial/legal liability	Stripe handles card data (PCI-DSS), no card storage
DDoS attack	Service unavailable	Cloud WAF + DDoS protection + rate limiting
Token theft	Account compromise	Short-lived JWTs (15m), refresh token rotation

End of High Level Design Document